University of Calgary
Faculty of Science
Final Exam PHYSICS 365

December 9, 2023, 1:00 PM- 4:00 PM Time: 3 hours

This is a closed-book exam containing a total of 30 equally weighted questions. Please answer all questions.
ONLY THE FOLLOWING ITEMS ARE ALLOWED ON YOUR DESK DURING THE EXAM:

1. **THIS MULTIPLE-CHOICE QUESTION BOOKLET**, which includes the multiple-choice exam questions and the formula sheet (last two pages). **THIS BOOKLET WILL BE HANDED IN**.

2. A BUBBLE SHEET used to answer multiple-choice questions. **IT WILL BE HANDED IN**.

IMPORTANT: START by entering your UCID NUMBER, NAME, and COURSE ID on the bubble sheet.

3. A CALCULATOR. Any scientific or graphing calculator (including a programmable one) may be used during the exam.

4. A PEN OR PENCIL, used to black out your answers on the bubble sheet. If you are using a PEN, make sure it is BLACK or BLUE in order to ensure the scanner reads your answers properly. PENCILS can also be used, but make sure you press firmly when answering so the scanner reads your answer properly.

5. YOUR STUDENT ID CARD, used to check your identity during exam sign-in.

6. Students who stay until 10 minutes before the end of the exam must remain seated until permitted to leave by the exam supervisor.

1. The velocity $v(t)$ of a particle undergoing SHM is graphed in Figure 1. At point A, the particle momentarily is:

   A) stationary
   B) headed toward $x = -x_m$
   C) headed toward $x = +x_m$
   D) headed toward $x = 0$

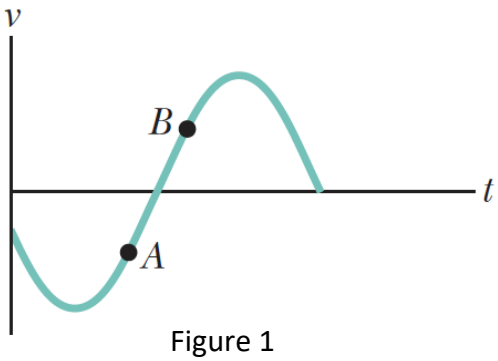


Figure 1

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2. What is the phase constant for the harmonic oscillator with the velocity function $v(t)$ given in Figure 2 if the position function $x(t)$ has the form $x = x_m \cos(\omega t + \phi)$? The vertical axis scale is set by v_s = 4.0 cm/s.

   A) 4.07 rad
   B) 3.25 rad
   C) 0.93 rad
   D) 5.36 rad

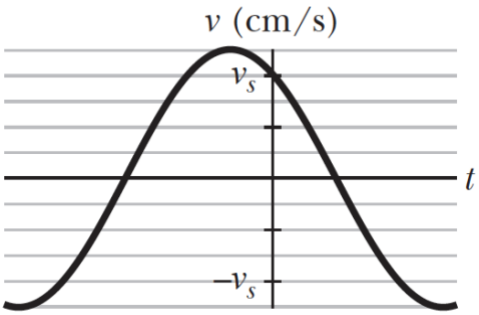


Figure 2

Questions 3, and 4 are related to the following situation:

You send a traveling wave along a particular string by oscillating one end.

3. If you increase the frequency of the oscillations, which statement is **true**?

   A) The speed of the wave remains the same.
   B) The speed of the wave increases.
   C) The wavelength of the wave remains the same.
   D) The wavelength of the wave increases.

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4. If, you increase the tension in the string, which statement is **true**?

   A) The speed of the wave decreases.
   B) The speed of the wave remains the same.
   C) The wavelength of the wave decreases.
   D) The wavelength of the wave increases.

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5. Two sinusoidal waves of the same frequency travel in the same direction along a string. If y_{m1} = 3.0 cm, y_{m2} = 5.0 cm, ϕ_1 = 0, and $\phi_2 = \pi/2$ rad, what is the amplitude of the resultant wave?

   A) 34 cm
   B) 0
   C) 15 cm
   D) 5.8 cm

6. Two very long strings are joined at the position $x = 0$ and placed under a tension of 100 N. The string at $x < 0$ has mass density $\mu = 0.5$ kg/m while the string at $x > 0$ has mass density $\mu = 1$ kg/m. A wave of the form $y(x,t) = A\cos(kx - \omega t)$ is incident from $x < 0$. If the incident angular frequency ω is 1.0 rad/s and the incident amplitude A is 10 cm, then the displacement of the particle at $x = +1.0$ m is:

A) $y(t) = (8.3\text{ cm})\cos(\omega t - 0.63)$
B) $y(t) = (6.7\text{ cm})\cos(\omega t - 0.63)$
C) $y(t) = (6.7\text{ cm})\cos(\omega t - 0.10)$
D) $y(t) = (8.3\ c\text{m})\cos(\omega t - 0.10)$

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7. In Figure 3, an electric spark jumps along a straight line of length L = 10 m, emitting a pulse of sound that travels radially outward from the spark. (The spark is said to be a line source of sound.) The power of this acoustic emission is P_s = 1.6 × 10^4 W. What is the intensity I of the sound when it reaches a distance r = 16 m from the spark?

A) 12 W/m^2
B) 10 W/m^2
C) 16 W/m^2
D) 21 W/m^2

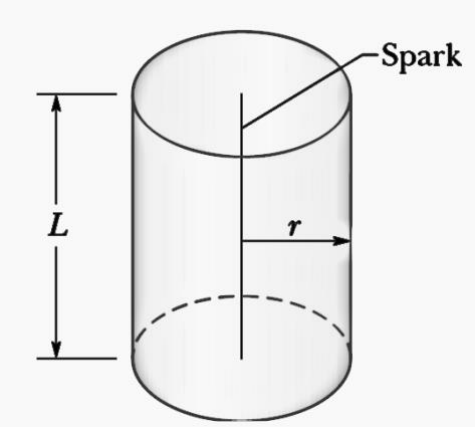


Figure 3

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8. Pipe A, with length L, and pipe B, with length $2L$, both have two open ends. Which harmonic of pipe B has the same frequency as the first harmonic (fundamental frequency) of pipe A?

   A) First
   B) Second
   C) Third
   D) Fourth

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9. For a particular tube with both ends closed, here are four of the six harmonic frequencies below 1000 Hz: 300 Hz, 600 Hz, 750 Hz, and 900 Hz. What two frequencies are missing from the list?

   A) 100 Hz, and 200 Hz
   B) 250 Hz, and 350 Hz
   C) 150 Hz, and 450 Hz
   D) 300 Hz, and 800 Hz

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10. Light from a HeNe laser (λ = 632.8 nm) is incident on 2 slits separated by 0.125 mm. Find the distance between the $m = +2$ and the $m = -2$ bright fringes on a screen 4.2 m away.

    A) 75 mm
    B) 65 mm
    C) 43 mm
    D) 85 mm

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11. A person on a railroad car blows a trumpet note at 440 Hz. The car is moving toward a wall at 20.0 m/s. What frequency does the trumpeter hear when the sound is reflected off the wall and back to the trumpeter?

A) 467 Hz
B) 494 Hz
C) 416 Hz
D) 440 Hz

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12. A guitar string has a mass per unit length of 0.5 g/m and a 65 cm length. What is the tension required to tune its fundamental frequency (first harmonic) to 330 Hz?

A) 74 N
B) 83 N
C) 92 N
D) 54 N

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13. Figure 4 shows two rays of light, of wavelength 600 nm, that reflect from glass surfaces separated by 150 nm. The rays are initially in phase. What is the path length difference of the rays?

A) 150 nm
B) 75 nm
C) 450 nm
D) 300 nm

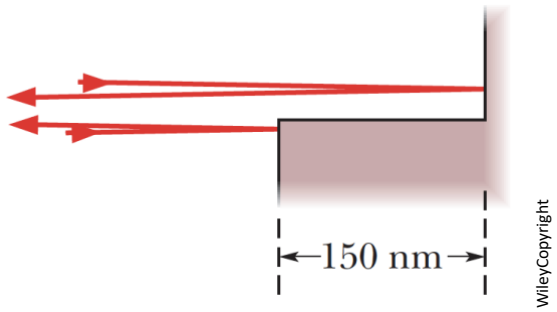


Figure 4

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14. In figure 5, a wave on a string is incident on a boundary between two strings. The wave travels from the left, on the string of linear mass density μ, to the right, onto the string of linear mass density $\alpha\mu$ (i.e., the linear mass density of the string on right is α times that of the string on the left). Which option gives the fraction of wave energy transmitted at this boundary?



Figure 5

A) $\frac{2\sqrt{\alpha}}{1+\sqrt{\alpha}}$
B) $\frac{4\sqrt{\alpha}}{(1+\sqrt{\alpha})^2}$
C) $\frac{2}{1+\sqrt{\alpha}}$
D) $\frac{4\sqrt{\alpha}}{1+\sqrt{\alpha}}$

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15. A slit 1.00 mm wide is illuminated by light of wavelength 589 nm. We see a diffraction pattern on a screen 3.00 m away. What is the distance between the first two diffraction minima on the same side of the central diffraction maximum?

A) 1.77 mm
B) 3.53 mm
C) 1.18 mm
D) 5.89 mm

Questions 16 and 17 are related to the following situation:

In the Figure 6, two isotropic point sources S_1 and S_2 emit light in phase at wavelength λ and at the same amplitude. The sources are separated by distance $d = 10\,\lambda$ on an x axis. A viewing screen is at distance $D = 20\,\lambda$ from S_2 and parallel to the y axis. The figure shows two rays reaching point P on the screen, at height y_P.

16. If $y_p = d$, what is the path length difference between the two rays?

A) 9.3 λ
B) 5.8 λ
C) 7.5 λ
D) 6.4 λ

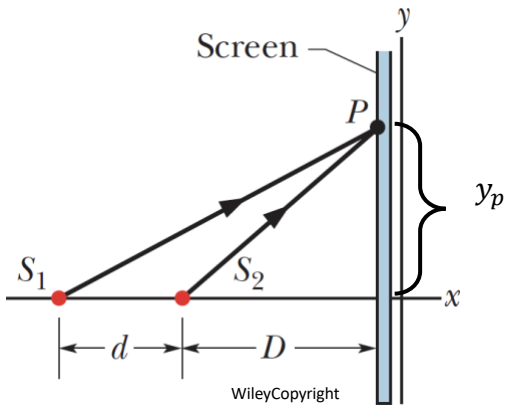


Figure 6

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17. If the path length difference between two rays from S_1 and S_2 at P (above figure) equals to 7.8 λ, the resulting intensity at point P is:

A) Minimum
B) Maximum
C) Intermediate but closer to maximum
D) Path length difference 7.8 λ for the above situation is impossible.

18. In Figure 7 , two springs are joined and connected to a block of mass 0.245 kg that is set oscillating over a frictionless floor. The springs each have spring constant k = 6430 N/m. What is the frequency of the oscillations?

A) 75 Hz
B) 54 Hz
C) 18 Hz
D) 38 Hz

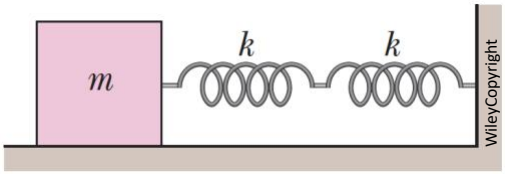


Figure 7

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19. A solar panel on a satellite having a surface area of 2.00 m^2 absorbs solar energy having an irradiance of 1360 W/m^2 and a wavelength of 500 nm. How many photons are absorbed each second, assuming 100% absorption?

A) 1.71×10^{21}
B) 3.42×10^{21}
C) 1.37×10^{22}
D) 6.84×10^{21}

20. In the bright and sunny Physics Lab #39458, photons are incident on a target containing stationary electrons. The incident radiation has a wavelength of 2.89×10^{-12} m, and the detector is placed at an angle of φ from the initial direction of the photons. An electron from which one of these photons was scattered receives an energy of 120.6 keV. What is the angle φ? (Answer in degrees, with $0 \leq \varphi \leq 180°$).

A) 57.6°
B) 62.4°
C) 122.4°
D) 117.6°

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21. You perform a pair of photoelectric experiments:

- Laser 1 is incident on surface 1, with work function Φ_1 = 2 eV.
- Laser 2 is incident on surface 2, with work function Φ_2 = 4 eV .

It is found that, in both cases, electrons with a maximum kinetic energy of 2 eV are ejected. How do the wavelengths of the lasers compare?

A) $\lambda_2 = \frac{1}{2}\lambda_1$
B) $\lambda_2 = \frac{3}{2}\lambda_1$
C) $\lambda_2 = \frac{2}{3}\lambda_1$
D) $\lambda_2 = 2\lambda_1$

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22. In an experimental set-up, three quantum tunneling conditions are tested. Electrons with energy E approach the following potential energy barriers. A detector placed on the other side measures if the electrons are transmitted.

- Barrier (1) has height of $10E$ and thickness of L.
- Barrier (2) has height of $17E$ and thickness of $L/2$.
- Barrier (3) has height of $2E$ and thickness of $2L$.

Rank the barriers according to the probability of the electron tunneling through the barrier, from the largest probability to smallest.

A) (2) = (3) > (1)
B) (1) > (2) = (3)
C) (1) > (3) > (2)
D) (2) > (3) > (1)

23**.** An infinite, one-dimensional potential well of width L contains a single electron of energy E. In repeated experiments it is found that there are four places within the well (including the two endpoints) where the electron is never detected. What is the electron energy E in terms of the ground state energy E_1?

A) E_1
B) 4 E_1
C) 9 E_1
D) 16 E_1

24. A one-dimensional infinite potential well in the region $0 \leq x \leq L$ has allowed electron energy levels of 1, 4, 9, 16, ... eV. If the width of the well is slowly doubled to $2L$, what is the new energy of the $n = 4$ level?

A) 1 eV
B) 2 eV
C) 4 eV
D) 8 eV

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25. A wavefunction describing a quantum particle in one dimension is given by

$$\psi(x) = \begin{cases} A\sqrt{x}, & \text{for } 0 \leq x \leq L \\ 0, & \text{for all other } x \end{cases}$$

If $L = 0.78$ mm, what value of A is needed to normalize this wavefunction?

A) $1.81\ \text{mm}^{-1}$

B) $2.56\ \text{mm}^{-1}$

C) $2.18\ \text{mm}^{-1}$

D) $0.30\ \text{mm}^{-1}$

26. An electron is in a one-dimensional infinite potential well of length L = 104 pm. The electron has ground state energy E_1. What is the longest wavelength photon that could be emitted by the system in a transition from the energy level $n = 3$?

A) 5.10 nm
B) 7.13 nm
C) 11.9 nm
D) 4.45 nm

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27. The energy levels of a hydrogen-like quantum system (with one electron bound to a potential well) are found to be

$$E_n = -2.6\text{eV}/n^2.$$

As the electron makes a transition from a bound state with quantum number n_{high} to a bound state with quantum number n_{low}, you observe a photon of wavelength 637 nm emitted from this system. What is n_{low}?

A) 2
B) 3
C) 4
D) 1

28. Imagine playing baseball in a universe (not ours!) where the Planck constant is 0.40 J·s. What would be the uncertainty in the position of a 0.40 kg baseball that is moving at 25 m/s along an axis if the uncertainty in the speed is 1.5 m/s?

A) 0.60 m
B) 0.11 m
C) 0.15 m
D) 0.30 m

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29. Light of wavelength 2.40 pm is directed onto a target containing free electrons. Find the wavelength of light scattered at 30° from the incident direction. The electron Compton wavelength is 2.43×10^{-12} m.

A) 3.65 pm
B) 6.04 pm
C) 2.73 pm
D) 4.32 pm

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30. Rank the following pairs of quantum states for an electron confined to an infinite well according to the energy differences between the states, greatest first: (a) n = 3 and n = 1, (b) n = 5 and n = 4, (c) n = 4 and n = 3.

A) (a) > (b) > (c)
B) (b) > (c) > (a)
C) (b) > (a) > (c)
D) (a) > (c) > (b)

Formula sheet PHYS 365

OSCILLATIONS			
$\omega = 2\pi f = \frac{2\pi}{T}$	$\omega = \sqrt{\frac{k}{m}}$	$\omega = \sqrt{\frac{g}{L}}$	$T = 2\pi\sqrt{\frac{l}{g}}$
WAVES			
$y(x,t) = f(x \pm vt)$	$y(x,t) = y_m \sin(kx \pm \omega t + \phi)$		
$v = \frac{\omega}{k} = f\lambda$	$k = \frac{2\pi}{\lambda}$	$v = \frac{c}{n}$	$\lambda = \frac{\lambda_{vacuuum}}{n}$
$v = \sqrt{\frac{\tau}{\mu}}$	$v = \sqrt{\frac{\gamma RT}{M}}$	$v = \sqrt{\frac{Y}{\rho}}$	$v = \sqrt{\frac{B}{\rho}}$
$Z = \sqrt{\mu\tau} = \frac{\tau}{v} = \mu v$	$Z = \sqrt{\rho B} = \rho v$	$P_{avg} = \frac{1}{2}\mu v\omega^2 y_m^2$	$\Delta p_m = (v\rho\omega)s_m$
$\Delta p = \Delta p_m \sin(kx - \omega t)$	$I = \frac{1}{2}\rho v\omega^2 s_m^2$	$I = \frac{P}{4\pi R^2}$	$I = \frac{P}{A}$
$\beta = (10\ dB)\log_{10}\left(\frac{I}{I_0}\right)$	$\Delta\beta = (10\ dB)\log_{10}\left(\frac{I_2}{I_1}\right)$	$I(r_2) = \left(\frac{r_1}{r_2}\right)^{n-1} I(r_1)e^{-\alpha\Delta r}$	
$R = \frac{Z_1 - Z_2}{Z_1 + Z_2}$	$T = \frac{2Z_1}{Z_1 + Z_2}$	$R^2 + \frac{Z_2}{Z_1}T^2 = 1$	$Y = \frac{\left(\frac{F}{A}\right)}{\left(\frac{\Delta L}{L}\right)}$
$f = \frac{v}{\lambda} = nf_1$	$\frac{\partial^2 y}{\partial^2 t} = v^2\frac{\partial^2 y}{\partial x^2}$	$\Delta p = -B\frac{\partial s}{\partial x}$	$B = -\frac{\Delta p}{\left(\frac{\Delta V}{V}\right)}$

$x(t) = x_m \cos(\omega t + \Phi)$		$s = s_m \cos(kx - \omega t)$
$y'(x,t) = \left[2y_m \cos\left(\frac{1}{2}\Phi\right)\right]\sin\left(kx - \omega t + \frac{1}{2}\Phi\right)$		$y'(x,t) = [2y_m \sin(kx)]\cos(\omega t)$
$\Phi = \frac{\Delta L}{\lambda}2\pi$	$f_{beat} = f_1 - f_2$	$f' = f\frac{v \pm v_D}{v \pm v_S}$
Constructive interference $\frac{\Delta L}{\lambda} = 0, 1, 2, \ldots$		Destructive interference $\frac{\Delta L}{\lambda} = 0.5, 1.5, 2.5, \ldots$
Pipe open at both ends $f = \frac{v}{\lambda} = \frac{nv}{2L}$, $n = 1, 2, 3, \ldots$		Pipe closed at one end and open at the other $f = \frac{v}{\lambda} = \frac{nv}{4L}$, $n = 1, 3, 5, \ldots$

$d\ \sin(\theta) = m\lambda$ for $m = 0, 1, 2, \ldots$ (Maxima- bright fringes)		
$d\ \sin(\theta) = \left(m + \frac{1}{2}\right)\lambda$ for $m = 0, 1, 2, \ldots$ (minima—dark fringes),		
$I = 4I_0 \cos^2\left(\frac{1}{2}\Phi\right)$	$\Phi = \frac{2\pi d}{\lambda}\sin(\theta)$	$I = I_0 \cos^2(\theta)$

Reflection	Reflection phase shift
Off lower index	0
Off higher index	0.5 wavelength

$a \sin(\theta) = m\lambda$ for $m = 1, 2, 3, \ldots$. (minima)

CONSTANTS

$c = 3.00 \times 10^8 \frac{m}{s}$	$R = 8.314 \frac{J}{k.mol}$	$M_{air} = 28.8 \frac{g}{mol}$	$g = 9.81 \frac{m}{s}$
$v_{sound\ at\ 20℃ = 343 \frac{m}{s}}$	$T_K = T_℃ + 273.15$	$\gamma_{monoatomic} = \frac{5}{3}$	$R_e = R^2$
$I_0 = 1 \times 10^{-12} \frac{W}{m^2}$	$1\ atm = 1.01 \times 10^5 Pa$	$\gamma_{diaatomic} = \frac{7}{5}$	$T_e = \frac{Z_2}{Z_1} T^2$

MATHEMATICAL FORMULAE

$\cos(\alpha) \pm \cos(\beta) = 2\cos\left(\frac{1}{2}(\alpha \pm \beta)\right)\cos\left(\frac{1}{2}(\alpha \mp \beta)\right)$	$\sin(\alpha) \pm \sin(\beta) = 2\sin\frac{(\alpha \pm \beta)}{2}\cos\frac{(\alpha \mp \beta)}{2}$
$\cos(\alpha \pm \beta) = \cos(\alpha)\cos(\beta) \mp \sin(\alpha)\sin(\beta)$	$\tilde{A} = Ae^{i\theta} = A(\cos(\theta) + i\ \sin(\theta))$
$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$	$\sin(\alpha \pm \beta) = \sin(\alpha)\cos(\beta) \pm \cos(\alpha)\sin(\beta)$
$\sin(-\alpha) = -\sin(\alpha)$	$\cos\left(\alpha \pm \frac{\pi}{2}\right) = \mp \sin(\alpha)$
$\cos(-\alpha) = \cos(\alpha)$	$\sin\left(\alpha \pm \frac{\pi}{2}\right) = \pm \cos(\alpha)$

MODERN PHYSICS

$F_e = \frac{1}{4\pi\epsilon_0}\frac{e^2}{r^2}$	$U = \frac{1}{4\pi\epsilon_0}\frac{e^2}{r}$	$U = qV$	$K = \frac{1}{2}mv^2$
$E = hf$	$p = \frac{h}{\lambda}$	$hf = K_{max} + \Phi$	$a_c = \frac{v^2}{r}$
$\Delta\lambda = \frac{\mathrm{h}}{\mathrm{mc}}(1 - cos\phi)$	$\frac{1}{\lambda} = \mathrm{R}\left(\frac{1}{\mathrm{n}_{\mathrm{low}}^2} - \frac{1}{\mathrm{n}_{\mathrm{high}}^2}\right)$	$R = \frac{P}{hf}$	
$\Delta x \Delta p_x \geq \frac{h}{2\pi}$	$a = \epsilon_0\frac{n^2h^2}{\pi m_e e^2}$	$E_n = -\frac{me^4}{8\epsilon_0^2h^2n^2}$	$E_n = -13.6\frac{eV}{n^2}$
$E_n = \frac{h^2}{8mL^2}n^2$	$\psi_n(x) = \sqrt{\frac{2}{L}}sin\left(\frac{n\pi}{L}x\right)$	$T \approx e^{-2bL}$ $b = \sqrt{\{8\pi^2m(U_b - E)/h^2}$	
$P(r) = \frac{4}{a_0^3}r^2e^{-\frac{2r}{a_0}}$	$\psi(r) = \frac{1}{\sqrt{\pi}a^{3/2}}e^{-\frac{r}{a}}$	$\frac{d^2\Psi}{dx^2} + \frac{8\pi^2m}{h^2}\big(E - U(x)\big)\psi = 0$	
$\frac{d^2\psi}{dx^2} + k^2\psi = 0,$ with $\quad k = \frac{2\pi\sqrt{2m(E-U)}}{h}$			

MATHEMATICAL FORMULAE

$\int cos(ax) = \frac{sin(ax)}{a} + c$	$\int sin(ax) = -\frac{cos(ax)}{a} + c$
$\int sin^2(ax)dx = \frac{x}{2} - \frac{sin(2ax)}{4a} + c$	$\int cos^2(ax)dx = \frac{x}{2} + \frac{sin(2ax\}}{4a} + c$
$sin^2\theta = \frac{1}{2}\big(1 - cos(2\theta)\big)$	$cos^2\theta = \frac{1}{2}\big(1 + cos(2\theta)\big)$

CONSTANTS

$m = 9.109 \cdot 10^{-31}kg$	$m_p = 1.67 \cdot 10^{-27}kg$
$a_0 = 0.5292 \cdot 10^{-10}m$	$1u = 1.66054 \cdot 10^{-27}kg$
$\epsilon_0 = 8.854 \cdot 10^{-12}\frac{C^2}{Nm^2}$	$h = 6.63 \cdot 10^{-34}J \cdot s = 4.136 \cdot 10^{-15}eV \cdot s$
$\sigma = 5.67 \cdot 10^{-8}\frac{W}{m^2K^2}$	$e = 1.602 \cdot 10^{-19}C$
$k_B = 1.381 \cdot 10^{-23}\frac{J}{K} = 8.617 \cdot 10^{-5}\frac{eV}{K}$	Rydberg constant $R = 1.09737 \cdot 10^7m^{-1}$